

Supplementary Material S1

Cost signatures of the standard release models, and their symbolic and numerical verification

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This document supports the article *Marginal time-cost analysis of drug release*. It contains two things the main text asserts but does not have room to show. Section S1 derives the cost function of each standard release model from first principles, so that every entry in Table 5 of the article can be checked line by line. Section S3 sets out the symbolic and numerical verification of every quantitative claim made in the article, with each check classified as an exact identity or a numerical approximation carrying an error budget.

Equation, table and section numbers in this document are prefixed S. References to numbered sections, equations and tables without that prefix refer to the main article.

The verification of Section S3 is reproduced as an executable notebook, and the identifiability failure of the main text is demonstrated interactively, at <https://jadi255.github.io/marginal-time-cost>; both are available in the public repository <https://github.com/Jadi255/marginal-time-cost>.

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S1. Cost signatures of the standard models

Applying the transform analytically to each standard model yields a distinct cost function. Together these form a reference library: a cost curve measured from data may be compared against these signatures. Each derivation follows the same three steps — differentiate $Q(t)$ to obtain the rate; invert to obtain cost; substitute $t(Q)$ to express cost in terms of Q .

All results in this section were verified symbolically using a computer algebra system. Those involving fractional powers (Sections S1.5 and S1.6) were additionally verified numerically.

S1.1 Zero-order

Release proceeds at a constant rate, independent of the amount remaining.

$$Q = k_0 t \quad (S1)$$

$$(i) \quad \frac{dQ}{dt} = k_0 \quad (S2)$$

$$(ii) \quad c = \frac{1}{k_0} \quad (S3)$$

No substitution is needed, since no t remains in the expression.

$$c(Q) = \frac{1}{k_0} \quad \text{CONSTANT} \quad (S4)$$

Signature: flat. Every unit costs the same amount of time. This is the reference case in which 5 mg takes exactly five times as long as 1 mg.

S1.2 Korsmeyer–Peppas (the general power law)

An empirical power law in time, with fitted exponent n .

$$Q = k t^n \quad (S5)$$

$$(i) \quad \frac{dQ}{dt} = k n t^{n-1} \quad (S6)$$

$$(ii) \quad c = \frac{1}{k n t^{n-1}} = \frac{t^{1-n}}{k n} \quad (S7)$$

$$(iii) \quad Q = k t^n \implies t = \left(\frac{Q}{k} \right)^{1/n} \quad (S8)$$

Substituting into (ii):

$$c = \frac{[(Q/k)^{1/n}]^{1-n}}{k n} = \frac{(Q/k)^{(1-n)/n}}{k n} \quad (S9)$$

$$c(Q) = \frac{Q^{(1-n)/n}}{n k^{1/n}} \implies c(Q) \propto Q^{b-1}, \quad b \equiv \frac{1}{n} \quad \text{MONOMIAL — NO } Q_\infty \text{ TERM} \quad (S10)$$

since the exponent $(1 - n)/n = 1/n - 1 = b - 1$.

Two observations follow. First, Q_∞ appears nowhere in this expression, which is the structural blindness identified in Section 1 of the article. Second, b is *not an independent quantity*: $b = 1/n$ exactly. It is the Peppas exponent inverted. Section 4.10 of the article explains why b should therefore not be presented as the headline result of this method.

S1.3 Higuchi (a special case of the above)

Release proportional to the square root of time.

$$Q = k_H \sqrt{t} = k_H t^{1/2} \quad (\text{S11})$$

$$(i) \quad \frac{dQ}{dt} = \frac{k_H}{2\sqrt{t}} \quad (\text{S12})$$

$$(ii) \quad c = \frac{2\sqrt{t}}{k_H} \quad (\text{S13})$$

$$(iii) \quad \sqrt{t} = \frac{Q}{k_H} \quad (\text{S14})$$

Substituting:

$$c = \frac{2(Q/k_H)}{k_H} \quad (\text{S15})$$

$$c(Q) = \frac{2Q}{k_H^2} \quad \text{LINEAR IN } Q \quad (\text{S16})$$

Consistency check. Setting $n = \frac{1}{2}$ and $k = k_H$ in the Peppas result of Section S1.2 gives exponent $(1 - \frac{1}{2})/\frac{1}{2} = 1$ and prefactor $1/(\frac{1}{2}k_H^2) = 2/k_H^2$, hence $c = 2Q/k_H^2$ — identical, as it must be, since Higuchi is Peppas with $n = \frac{1}{2}$. *Confirmed symbolically.*

Signature: rises linearly and remains finite everywhere. The marginal cost grows in direct proportion to the amount already released, which is the classical picture of a lengthening diffusion path.

S1.4 First-order

The rate is proportional to the amount of drug remaining.

$$Q = Q_\infty (1 - e^{-kt}) \quad (\text{S17})$$

$$(i) \quad \frac{dQ}{dt} = Q_\infty k e^{-kt} \quad (\text{S18})$$

A useful identity, obtained directly from the model:

$$Q_\infty - Q = Q_\infty e^{-kt} \quad \implies \quad e^{-kt} = \frac{Q_\infty - Q}{Q_\infty} \quad (\text{S19})$$

Substituting into (i):

$$\frac{dQ}{dt} = Q_\infty k \frac{Q_\infty - Q}{Q_\infty} = k(Q_\infty - Q) \quad (\text{S20})$$

$$c(Q) = \frac{1}{k(Q_\infty - Q)} \quad \text{DIVERGES AS } Q \rightarrow Q_\infty \quad (\text{S21})$$

Signature: nearly flat early, then rising without limit at the ceiling. The physical reading is exact: the rate is proportional to the drug remaining, so as the remainder vanishes the cost of the next unit becomes unbounded. The final molecule takes forever.

It is worth noting that this derivation is *simpler* in cost coordinates than in time coordinates. The exponential disappears entirely, and the intermediate result $dQ/dt = k(Q_\infty - Q)$ — “the rate is proportional to what is left” — is the definition of first-order kinetics, recovered in a single line.

S1.5 Hixson–Crowell (the cube-root law)

Derived for dissolution from shrinking particles. It is stated in terms of the amount remaining, $W = Q_\infty - Q$, and is most safely differentiated in that form.

$$W^{1/3} = Q_\infty^{1/3} - \kappa t \quad \text{where } W = Q_\infty - Q \quad (\text{S22})$$

Differentiating implicitly with respect to t :

$$\frac{1}{3} W^{-2/3} \frac{dW}{dt} = -\kappa \quad \implies \quad \frac{dW}{dt} = -3\kappa W^{2/3} \quad (\text{S23})$$

Since $Q = Q_\infty - W$, we have $dQ/dt = -dW/dt$:

$$\frac{dQ}{dt} = 3\kappa W^{2/3} = 3\kappa (Q_\infty - Q)^{2/3} \quad (\text{S24})$$

$$c(Q) = \frac{1}{3\kappa (Q_\infty - Q)^{2/3}} \quad \text{DIVERGES AS } Q \rightarrow Q_\infty \quad (\text{S25})$$

Numerical verification: central-difference agreement with the explicit solution at $t = 0.5, 3, 10, 25$ and 30 , with residuals at the estimator's round-off floor ($\sim 10^{-9}$); the error budget is set out in Appendix S3. Working in terms of W avoids a complex cube-root branch that a direct symbolic solution introduces.

Signature: diverges, but more gently than first-order (exponent $-2/3$ rather than -1). The physical reading is dissolution from a shrinking particle whose surface area falls as $W^{2/3}$.

S1.6 Weibull

A flexible empirical form, stated in terms of the fraction released.

$$f = 1 - \exp\left[-\left(\frac{t}{\tau}\right)^\beta\right] \quad (\text{S26})$$

Inverting:

$$1 - f = \exp\left[-\left(\frac{t}{\tau}\right)^\beta\right] \quad (\text{S27})$$

$$-\ln(1 - f) = \left(\frac{t}{\tau}\right)^\beta \quad (\text{S28})$$

$$t = \tau [-\ln(1 - f)]^{1/\beta} \quad (\text{S29})$$

Differentiating with respect to f :

$$\frac{dt}{df} = \frac{\tau}{\beta} \cdot \frac{[-\ln(1 - f)]^{\frac{1}{\beta}-1}}{1 - f}, \quad c(Q) = \frac{1}{Q_\infty} \frac{dt}{df} \quad (\text{S30})$$

Consistency check. At $\beta = 1$ this reduces to $dt/df = \tau/(1 - f)$, which with $\tau = 1/k$ is exactly the first-order result of Section S1.4. *Confirmed symbolically.* Weibull contains first-order as a special case, just as Peppas contains Higuchi.

Signature: diverges at the ceiling, driven by the $1/(1 - f)$ factor.

S1.7 The library assembled

S2. The two families and the terminal invariant

The library divides cleanly into two groups.

S2.1 Family I — non-depleting power laws

Zero-order, Higuchi and Korsmeyer–Peppas. The cost is a monomial in Q . No Q_∞ appears anywhere, and the cost never diverges. These models describe a source that does not know it is running out. Members are distinguished from one another by a single exponent.

Model	$Q(t)$	$c(Q) = dt/dQ$	Shape	Ceiling?
Zero-order	$k_0 t$	$\frac{1}{k_0}$	flat	no
Peppas	$k t^n$	$\propto Q^{b-1}, b = 1/n$	monomial	no
Higuchi	$k_H \sqrt{t}$	$\frac{2Q}{k_H^2}$	linear rise	no
First-order	$Q_\infty(1 - e^{-kt})$	$\frac{1}{k(Q_\infty - Q)}$	diverges (-1)	yes
Hixson–Crowell	cube-root law	$\frac{1}{3\kappa(Q_\infty - Q)^{2/3}}$	diverges ($-2/3$)	yes
Weibull	$1 - e^{-(t/\tau)^\beta}$	$\frac{\tau}{\beta Q_\infty} \frac{[-\ln(1-f)]^{\frac{1}{\beta}-1}}{1-f}$	diverges	yes

Table S1: The cost signature of each standard model. The presence or absence of a Q_∞ term determines the family.

S2.2 Family II — depleting or ceiling laws

First-order, Hixson–Crowell and Weibull. The cost *diverges* as $Q \rightarrow Q_\infty$, because the driving force — the drug remaining, or the available surface area — vanishes. Members are distinguished not by a single exponent but by the *manner* of their divergence: the power of $(Q_\infty - Q)$ is -1 for first-order and $-2/3$ for Hixson–Crowell.

S2.3 The discriminating invariant

These two families are nearly indistinguishable in cumulative coordinates, as Section 1 of the article demonstrated, yet qualitatively different in cost coordinates. This motivates the central general statement of the method:

The behaviour of $c(Q)$ as $Q \rightarrow Q_\infty$ is the primary model-discriminating invariant.

The concentration of discriminating power in the terminal region can be quantified. Anchoring the Higuchi and first-order models at a common half-release time (T_{50}), so that they agree on the midpoint of the profile, their cost functions remain within a factor of 2 of one another across the entire range $f = 0.1$ to 0.9 — a difference indistinguishable given ordinary experimental error. At $f = 0.99$ they differ by a factor of 18.2. Discrimination is concentrated almost entirely at the end of the curve.

The argument now closes on itself. The terminal region is exactly what the Korsmeyer–Peppas convention discards, and Family I comprises exactly those models that possess no Q_∞ term. The conventional framework is built from the family that cannot represent a ceiling, and it truncates away the region in which ceilings become apparent.

S3. Numerical and symbolic verification

This appendix shows the work behind every verification claim made in the body of the report. Each check is classified as either an *exact identity* — established symbolically, where the residual is zero by construction — or a *numerical approximation*, where a finite residual is expected and its size is accounted for. Distinguishing these matters: reporting an exact identity as though it were an approximation understates it, and reporting an approximation without its error budget overstates it.

Symbolic work was performed with a computer algebra system; numerical work in double precision (machine epsilon $\varepsilon \approx 2.2 \times 10^{-16}$).

S3.1 Check 1 — Recovery of the release curve (exact)

Claim (Section 2.2 of the article). $t(Q) = \int_0^Q c(u) du$ recovers the release curve.

Work. For Higuchi, $c(u) = 2u/k_H^2$. Integrating:

$$\int_0^Q \frac{2u}{k_H^2} du = \frac{2}{k_H^2} \left[\frac{u^2}{2} \right]_0^Q = \frac{Q^2}{k_H^2} \quad (\text{S31})$$

Substituting the Higuchi profile itself, $Q = k_H \sqrt{t}$:

$$\frac{Q^2}{k_H^2} = \frac{(k_H \sqrt{t})^2}{k_H^2} = \frac{k_H^2 t}{k_H^2} = t \quad (\text{S32})$$

The integral returns t *identically*, for every k_H and every t . This is an exact algebraic identity, not a numerical agreement.

Numerical confirmation ($k_H = 12$, trapezoidal rule, 2×10^5 intervals). Note that $c(Q)$ is *linear* in Q , and the trapezoidal rule integrates linear functions exactly; therefore the only possible error is floating-point round-off, and the observed residual is exactly zero:

t (min)	$Q = k_H \sqrt{t}$	$\int_0^Q c du$ (trapezoid)	residual $ t - \int $
1.0	12.000000	1.000000000000	0.0e + 00
4.0	24.000000	4.000000000000	0.0e + 00
9.0	36.000000	9.000000000000	0.0e + 00

Table S2: Recovery is exact. The residual is zero, not merely small.

S3.2 Check 2 — Higuchi is the $n = \frac{1}{2}$ case of Peppas (exact)

Claim (Section S1.3). Setting $n = \frac{1}{2}$ in the Peppas cost function returns the Higuchi cost function.

Work. The general Peppas result is $c(Q) = Q^{(1-n)/n} / (n k^{1/n})$. Substituting $n = \frac{1}{2}$, $k = k_H$ term by term:

$$\text{exponent: } \frac{1-n}{n} \Big|_{n=1/2} = \frac{1-\frac{1}{2}}{\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1 \quad (\text{S33})$$

$$\text{prefactor: } n k^{1/n} \Big|_{n=1/2, k=k_H} = \frac{1}{2} \cdot k_H^2 = \frac{k_H^2}{2} \quad (\text{S34})$$

$$\implies c(Q) = \frac{Q^1}{k_H^2/2} = \frac{2Q}{k_H^2} \quad (\text{S35})$$

which is the Higuchi cost function derived independently in Section S1.3. Symbolic subtraction of the two expressions returns exactly 0.

S3.3 Check 3 — Hixson–Crowell derivative (numerical, with error budget)

Claim (Section S1.5). $dQ/dt = 3\kappa(Q_\infty - Q)^{2/3}$.

Work. This is the one check that is genuinely numerical, because it tests an implicit differentiation against the explicit solution $Q(t) = Q_\infty - (Q_\infty^{1/3} - \kappa t)^3$. The derivative is estimated by central difference, $[Q(t+h) - Q(t-h)]/(2h)$ with $h = 10^{-6}$, and compared against the analytic claim. Parameters: $Q_\infty = 100$, $\kappa = 0.15$ (release completes at $t = 30.94$ min).

t (min)	$Q(t)$ (%)	central difference	$3\kappa(Q_\infty - Q)^{2/3}$	abs. residual
0.5	4.76957	9.384180110	9.384180109	$1.44e-09$
3.0	26.35623	7.906237627	7.906237628	$2.77e-10$
10.0	68.99384	4.441311184	4.441311180	$3.95e-09$
25.0	99.29125	0.357718790	0.357718792	$2.00e-09$
30.0	99.99716	0.009021328	0.009021329	$1.33e-09$

Table S3: Residuals sit at the round-off floor of the central-difference estimator.

Error budget. The residuals ($\sim 10^{-9}$) are exactly what theory predicts and are *not* evidence of a discrepancy in the claim. A central difference carries two error terms: truncation, of order $h^2 = 10^{-12}$, and round-off, of order $\varepsilon/h = 2.2 \times 10^{-16}/10^{-6} \approx 2 \times 10^{-10}$. The observed residuals are the sum of these, so the analytic expression is confirmed to the full precision the estimator can offer.

S3.4 Check 4 — Weibull reduces to first-order at $\beta = 1$ (exact)

Claim (Section S1.6). At $\beta = 1$ the Weibull cost function becomes the first-order cost function.

Work. Begin from the Weibull result and set $\beta = 1$. The logarithmic factor collapses because any positive quantity raised to the power zero is unity:

$$\frac{dt}{df} = \frac{\tau}{\beta} \cdot \frac{[-\ln(1-f)]^{\frac{1}{\beta}-1}}{1-f} \xrightarrow{\beta=1} \frac{\tau}{1} \cdot \frac{[-\ln(1-f)]^0}{1-f} = \frac{\tau}{1-f} \quad (\text{S36})$$

Convert to a cost per unit amount, and substitute $\tau = 1/k$ together with $f = Q/Q_\infty$, so that $1-f = (Q_\infty - Q)/Q_\infty$:

$$c(Q) = \frac{1}{Q_\infty} \frac{dt}{df} = \frac{1}{Q_\infty} \cdot \frac{1/k}{(Q_\infty - Q)/Q_\infty} = \frac{1}{Q_\infty} \cdot \frac{Q_\infty}{k(Q_\infty - Q)} = \frac{1}{k(Q_\infty - Q)} \quad (\text{S37})$$

which is exactly the first-order cost function of Section S1.4. Symbolic subtraction returns 0.

S3.5 Check 5 — Marginal-to-average ratio (exact)

Claim (Section 2.7 of the article). For a power law $t = a Q^b$, the ratio of marginal to average cost equals b .

Work. Both quantities carry the same factor $a Q^{b-1}$, which cancels:

$$\text{marginal: } c(Q) = \frac{d}{dQ}(a Q^b) = a b Q^{b-1} \quad (\text{S38})$$

$$\text{average: } \bar{a}(Q) = \frac{a Q^b}{Q} = a Q^{b-1} \quad (\text{S39})$$

$$\text{ratio: } \frac{c(Q)}{\bar{a}(Q)} = \frac{a b Q^{b-1}}{a Q^{b-1}} = b \quad (\text{S40})$$

The cancellation is complete, so the ratio is independent of Q — it is b everywhere, not merely on average. Numerical spot-check with $a = 0.35$, $b = 1.4$:

Q	marginal abQ^{b-1}	average aQ^{b-1}	ratio	ratio - b
10	1.230824351	0.879160251	1.400000000	$0.0e + 00$
40	2.142989665	1.530706904	1.400000000	$2.2e - 16$
80	2.827691816	2.019779868	1.400000000	$0.0e + 00$

Table S4: The ratio is b at every Q , to machine precision.

S3.6 Check 6 — The mean dissolution time identity (exact, and general)

Claim (Section 4.7 of the article). $MDT = \frac{1}{Q_\infty} \int_0^{Q_\infty} (Q_\infty - u) c(u) du$.

The body of the report verified this at two points. The integral in fact solves in closed form for the whole Family I, which subsumes both spot-checks and yields a result worth stating in its own right.

General work. For any power law $t = a Q^b$, the cost is $c(u) = a b u^{b-1}$. Then:

$$MDT = \frac{1}{Q_\infty} \int_0^{Q_\infty} (Q_\infty - u) a b u^{b-1} du \quad (S41)$$

$$= \frac{a b}{Q_\infty} \int_0^{Q_\infty} (Q_\infty u^{b-1} - u^b) du \quad (S42)$$

$$= \frac{a b}{Q_\infty} \left[Q_\infty \frac{u^b}{b} - \frac{u^{b+1}}{b+1} \right]_0^{Q_\infty} = \frac{a b}{Q_\infty} Q_\infty^{b+1} \left(\frac{1}{b} - \frac{1}{b+1} \right) \quad (S43)$$

$$= a b Q_\infty^b \cdot \frac{1}{b(b+1)} = \frac{a Q_\infty^b}{b+1} \quad (S44)$$

Since the total release time is $T_\infty = t(Q_\infty) = a Q_\infty^b$, this gives a compact and apparently under-appreciated result:

$$\frac{MDT}{T_\infty} = \frac{1}{b+1} \quad \text{exactly, for any Family I power law — independent of } a \text{ and } Q_\infty. \quad (S45)$$

Both numerical checks reported in Section 4.7 of the article are corollaries of Equation (S45):

- Zero-order has $b = 1$, so $MDT/T_\infty = 1/2$.
- Higuchi has $b = 2$, so $MDT/T_\infty = 1/3$.

Zero-order worked explicitly ($k_0 = 2$, $Q_\infty = 100$, so $c = 1/k_0 = \frac{1}{2}$ and $t(Q) = Q/2$). Both routes must agree:

$$\text{weighted form: } \frac{1}{100} \int_0^{100} (100 - u) \cdot \frac{1}{2} du = \frac{1}{200} \left[100u - \frac{u^2}{2} \right]_0^{100} = \frac{10000 - 5000}{200} = 25 \quad (S46)$$

$$\text{direct definition: } \frac{1}{100} \int_0^{100} \frac{u}{2} du = \frac{1}{100} \left[\frac{u^2}{4} \right]_0^{100} = \frac{2500}{100} = 25 \quad (S47)$$

$$\text{theory: } \frac{T_\infty}{2} = \frac{Q_\infty/k_0}{2} = \frac{50}{2} = 25 \quad (S48)$$

Higuchi worked explicitly ($k_H = 12$, $Q_\infty = 100$, so $c = 2Q/144$ and $t(Q) = Q^2/144$):

$$\text{weighted form: } \frac{1}{100} \int_0^{100} (100 - u) \frac{2u}{144} du = \frac{2}{14400} \left[50u^2 - \frac{u^3}{3} \right]_0^{100} = \frac{625}{27} = 23.148148 \dots \quad (\text{S49})$$

$$\text{direct definition: } \frac{1}{100} \int_0^{100} \frac{u^2}{144} du = \frac{1}{14400} \left[\frac{u^3}{3} \right]_0^{100} = \frac{625}{27} = 23.148148 \dots \quad (\text{S50})$$

$$T_\infty = \frac{Q_\infty^2}{k_H^2} = \frac{10000}{144} = \frac{625}{9} = 69.444 \dots \quad (\text{S51})$$

$$\frac{\text{MDT}}{T_\infty} = \frac{625/27}{625/9} = \frac{1}{3} \quad (\text{S52})$$

Both routes return the exact rational $625/27$, and the ratio is exactly $\frac{1}{3}$ as Equation (S45) requires with $b = 2$.

A consequence worth noting. Equation (S45) inverts to $b = T_\infty/\text{MDT} - 1$. Where a power law holds, the cost exponent is therefore recoverable from two integral quantities the field already reports, with no regression of any kind. This is a second route to b alongside the marginal-to-average ratio of Section 2.7 of the article, and it inherits the same caveat: it presumes a power law rather than demonstrating one, and it is confined to Family I, since T_∞ and MDT both diverge or lose meaning where the cost diverges.

S3.7 Check 7 — Tail-share values (exact)

Claim (Section 4.8 of the article). The tabulated values of $1 - T_{90}/T_{100}$.

Work. For a power law, $T_X = \int_0^{X\% \cdot Q_\infty} c dQ = a (X\% \cdot Q_\infty)^b$, so the constant a and the ceiling Q_∞ cancel in the ratio:

$$\frac{T_{90}}{T_{100}} = \frac{a (0.9 Q_\infty)^b}{a Q_\infty^b} = 0.9^b \quad \implies \quad 1 - \frac{T_{90}}{T_{100}} = 1 - 0.9^b \quad (\text{S53})$$

Release behaviour	b	0.9^b	$1 - 0.9^b$
Zero-order	1.0000	0.900000	0.100000
Peppas, $n = 0.7$	1.4286	0.860265	0.139735
Higuchi	2.0000	0.810000	0.190000

Table S5: Arithmetic behind the tail-share values.

S3.8 Summary of verification status

A correction to an earlier statement of this work. Previous drafts described Check 1 as agreeing “to six decimal places”. That wording was inaccurate in a way that understated the result: it implied a numerical approximation, whereas Section S3.1 shows the recovery is an exact algebraic identity whose numerical residual is precisely zero. The distinction is not pedantic — an exact identity and a close numerical agreement are different kinds of evidence, and conflating them invites a reviewer to suspect that the other checks are also softer than claimed.

Check	Status	Residual
1. Recovery $t(Q) = \int_0^Q c \, du$	Exact identity	0 (analytic and numerical)
2. Higuchi = Peppas at $n = \frac{1}{2}$	Exact identity	0 (symbolic)
3. Hixson–Crowell derivative	Numerical	$\sim 10^{-9}$ (round-off floor)
4. Weibull $\rightarrow$ first-order at $\beta = 1$	Exact identity	0 (symbolic)
5. Marginal/average ratio = b	Exact identity	$\leq 2.2 \times 10^{-16}$
6. MDT weighted-integral identity	Exact identity	0 (exact rationals)
7. Tail-share values	Exact arithmetic	0

Table S6: Six of the seven checks are exact identities with zero residual. Only Check 3 is a genuine numerical approximation, and its residual sits at the estimator’s round-off floor.